

EDUCATION

Swarthmore College, Philadelphia, PA

Aug 2023—May 2027

B.S. General Engineering (Electrical focus) & B.A. Applied Mathematics | GPA: 3.98/4.00

Teaching Assistant for Electrical Circuit Analysis (Aug 2025—Jan 2026) | NCAA Varsity Soccer (Aug 2023—Jun 2026)

Relevant Coursework: Electronic Circuit Applications, Digital Signal Processing, Electrodynamics, Computer Vision, Data Structures & Algorithms, Partial Differential Equations, Numerical PDEs, Thermo-Fluid Mechanics, Statistical Signal Processing

RESEARCH AND WORK EXPERIENCE

- Plasma Synthesis and Materials Characterization Research Intern

May 2026—Aug 2026

US Naval Research Laboratory

Washington, DC

- Developed inductively coupled plasma (ICP) reactor to synthesize nanoparticles; conducted plasma experiments.
 - Minimized reflected power in ICP by tuning the RF matching network using Smith chart; mitigated RF EMI.
 - Designed and ran atomic-layer-deposition (ALD) experiments to determine how dose parameters affect ALD coatings.
 - Used scanning electron microscopy & energy-dispersive spectroscopy to characterize ALD nanomaterial compacts.
- Advanced Materials Intern & Research Assistant

May 2025—Aug 2025

US Naval Research Laboratory

Washington, DC

- Measured thermal diffusivity and specific heat of thermoelectric semiconductors using laser flash analysis (LFA).
 - Designed and built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring to automate LFA experiments, including a GUI and data-logging interface to the lab computer.
 - Developed experiments to improve the lab’s measurement procedure for IR-transparent and high-resistance samples.
- Applied Mathematics Research Assistant

Jan 2024—Present

Swarthmore College NaSCaRS Lab

Philadelphia, PA

- Utilizing principles of computational fluid dynamics and numerical methods to develop high-order accurate PDE solvers for kinetic & fluid plasma models in MATLAB.
 - Co-authored [paper](#) on low-rank, structure-preserving solver for Vlasov-Fokker-Planck kinetic-ion fluid-electron model.
 - Presented at 2025 SIAM NNP conference, 2024 & 2025 *Sigma Xi* Swarthmore poster sessions.
- Electrical Engineering Research Assistant

Nov 2023—May 2024

Swarthmore College Engineering

Philadelphia, PA

- Used Arduino, MATLAB & ViscousFlow to simulate vortex-shedding & power output for a wind-energy harvester.
- Software Engineering Intern

Jun 2022—Aug 2022

Oregon Health and Science University

Portland, OR

- Used Kotlin to develop Android app that detects heart murmurs using audio data via Bluetooth stethoscope.

PROJECTS

- Convolutional Neural Network for Hand-Drawn Image Classification: [project link](#)

May 2026

- Designed and trained a CNN to classify 250 object categories, deployed it online for real-time evaluation using ONNX.
- Analog Circuit and PCB Design – Fusion360

Aug 2025—Dec 2025

- Designed and fabricated PCBs combining MOSFETs, sensors, actuators, op-amps, comparators, and audio amplifiers across four projects: [audio equalizer](#), [temp-controlled fan](#), [light-optimizing solar panel](#), [regulated DC power supply](#).
- AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, [link](#)

Feb 2025
- Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, [link](#)

Apr 2022—Feb 2023

TECHNICAL SKILLS

Technical: SPICE, PCB Design, CAD, Fusion360, Soldering, Embedded systems, PDE solvers, Numerical methods

Programming/Software: Python, MATLAB, Arduino, C++, LaTeX, Git, SolidWorks, AutoCAD, TensorFlow, PyTorch, OpenCV

Languages: Spanish (Fluent), Global Seal of Biliteracy (2022) | Citizenship: USA, Ireland

AWARDS & HONORS

- Tau Beta Pi Engineering Honors Society

Mar 2026
- Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society

Feb 2025, Mar 2026
- Donna Prentice Memorial Scholarship, American Society of Civil Engineers

Feb 2024
- National Merit Scholarship, National Merit Scholarship Corporation

Apr 2023